

Employee Attrition Prediction – Project Summary

Project Overview

This project focuses on predicting employee attrition using machine learning techniques. The main objective is to identify employees who are likely to leave the organization based on factors such as age, monthly income, overtime, job role, work experience, and other HR-related attributes. Using Python and machine learning libraries, a predictive model was developed to help organizations improve employee retention and support data-driven HR decisions.

Methodology

The project began with data preprocessing, including removing unnecessary columns and converting categorical variables into numerical format using one-hot encoding. The dataset was divided into 80% training and 20% testing data. Four classification algorithms—Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting—were trained and evaluated. Model performance was measured using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC Curve.

Key Insights

Primary Drivers: Monthly Income, Overtime, Age, and Total Working Years were the most influential factors affecting employee attrition.

Secondary Factors: Distance from Home, Years at Company, Daily Rate, Monthly Rate, and Number of Companies Worked also contributed significantly to employee turnover.

Model Behavior: Random Forest and Gradient Boosting produced the most reliable predictions and effectively identified employees with a higher risk of leaving the organization.

Performance

Among the four machine learning models, Random Forest and Gradient Boosting achieved the highest prediction accuracy of approximately 87% , outperforming Logistic Regression and Decision Tree. Feature importance analysis further confirmed that salary, overtime, and work experience were the strongest predictors of employee attrition.

Conclusion

This project demonstrated how machine learning can be effectively used to predict employee attrition and identify the key factors influencing employee turnover. The developed model provides valuable insights for HR departments, enabling proactive retention strategies, reducing recruitment costs, and improving workforce management.